

Class 19 Mini Project

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Table of contents

Investigating pertussis cases by year	1
A tale of two vaccines (wP & aP)	2
Exploring CMI-PB data	4
Merging tables	8

Investigating pertussis cases by year

Pertussis is a bacterial lung infection also known as Whooping Cough. Let's begin by examining CDC reported case numbers in the U.S.:

Q1. With the help of the R “addin” package datapasta assign the CDC pertussis case number data to a data frame called `cdc` and use `ggplot` to make a plot of cases numbers over time.

```
cdc<-data.frame(  
  year = c(1922L,1923L,1924L,1925L,1926L,  
            1927L,1928L,1929L,1930L,1931L,1932L,  
            1933L,1934L,1935L,1936L,1937L,1938L,  
            1939L,1940L,1941L,1942L,1943L,  
            1944L,1945L,1946L,1947L,1948L,1949L,  
            1950L,1951L,1952L,1953L,1954L,1955L,  
            1956L,1957L,1958L,1959L,1960L,  
            1961L,1962L,1963L,1964L,1965L,1966L,  
            1967L,1968L,1969L,1970L,1971L,1972L,  
            1973L,1974L,1975L,1976L,1977L,1978L,  
            1979L,1980L,1981L,1982L,1983L,  
            1984L,1985L,1986L,1987L,1988L,1989L,  
            1990L,1991L,1992L,1993L,1994L,1995L,  
            1996L,1997L,1998L,1999L,2000L,
```

```

2001L,2002L,2003L,2004L,2005L,2006L,
2007L,2008L,2009L,2010L,2011L,2012L,
2013L,2014L,2015L,2016L,2017L,2018L,
2019L,2020L,2021L,2022L,2023L,2024L),
cases = c(107473,164191,165418,152003,
202210,181411,161799,197371,166914,
172559,215343,179135,265269,180518,
147237,214652,227319,103188,183866,
222202,191383,191890,109873,133792,
109860,156517,74715,69479,120718,68687,
45030,37129,60886,62786,31732,28295,
32148,40005,14809,11468,17749,
17135,13005,6799,7717,9718,4810,3285,
4249,3036,3287,1759,2402,1738,
1010,2177,2063,1623,1730,1248,1895,
2463,2276,3589,4195,2823,3450,4157,
4570,2719,4083,6586,4617,5137,
7796,6564,7405,7298,7867,7580,9771,
11647,25827,25616,15632,10454,13278,
16858,27550,18719,48277,28639,
32971,20762,17972,18975,15609,18617,
6124,2116,3044,7063,22538)
)

```

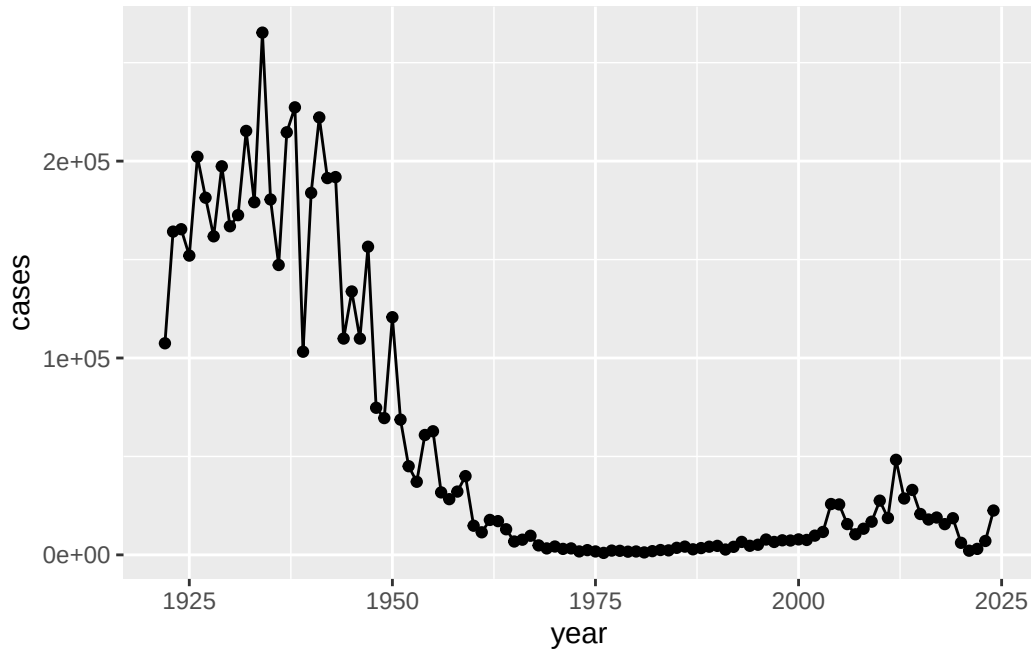
A tale of two vaccines (wP & aP)

Plot of cases per year for Pertussis in the U.S.

```

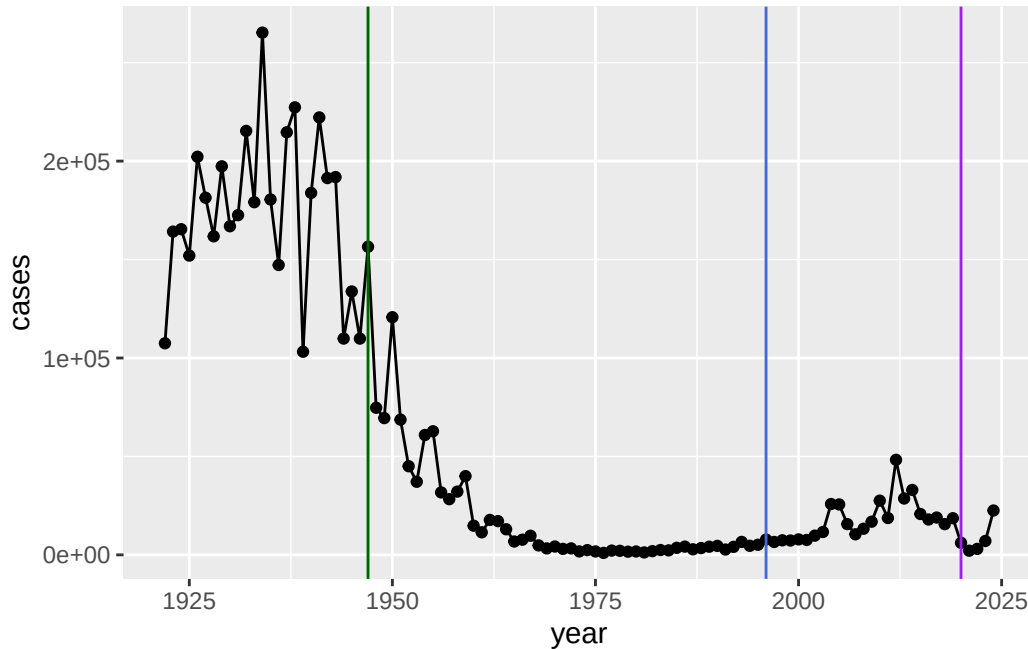
library(ggplot2)
ggplot(cdc)+
  aes(year,cases)+
  geom_point()+
  geom_line()

```



Add some major milestone timepoints to our plot: >Q2. Using the ggplot geom_vline() function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice? **Answer:** After switching, there is an increase in cases.

```
library(ggplot2)
ggplot(cdc)+
  aes(year,cases)+
  geom_point()+
  geom_line()+
  geom_vline(xintercept = 1947,col="darkgreen")+
  geom_vline(xintercept = 1996,col="royalblue")+
  geom_vline(xintercept=2020,col="purple")
```



The full introduction of mandatory wP (whole-cell) Pertussis immunization in the mid 1940s lead to a dramatic reduction in case numbers (from over 200,000 to 100s).

The switch to the aP (newer acellular) formula created an increase in cases for many countries such as Japan.

Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

Answer: The formula could simply be not as effected or well formulated as the older one, less people could also be getting the vaccine due to anti-vax sentiment.

Exploring CMI-PB data

The mission of CMI-PB is to provide the scientific community with a comprehensive, high-quality and freely accessible resource of Pertussis booster vaccination.

The CMI-PD website: <https://www.cmi-pb.org/>

They make their data available via JSON format API endpoints - basically the database tables in a key:value format like "infancy_vas": "wP". To read this we can use the 'read_json()' function from the **jsonlite** package.

```
library(jsonlite)

subject <-read_json(path="http://cmi-pb.org/api/v5_1/subject",simplifyVector = TRUE)

head(subject)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	2	wP	Female	Not Hispanic or Latino	White
3	3	wP	Female	Unknown	White
4	4	wP	Male	Not Hispanic or Latino	Asian
5	5	wP	Male	Not Hispanic or Latino	Asian
6	6	wP	Female	Not Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset
4	1988-01-01	2016-08-29	2020_dataset
5	1991-01-01	2016-08-29	2020_dataset
6	1988-01-01	2016-10-10	2020_dataset

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

Answer: There are 87 aP and 85 wP with 172 total subjects.

```
nrow(subject)
```

```
[1] 172
```

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q5. How many Male and Female subjects/patients are in the dataset?

Answer: There are 112 females and 60 males.

```
table(subject$biological_sex)
```

```
Female  Male
112     60
```

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc...)?

Answer: There are more females than males and the majority of race is white and asian. It is not very representative of the U.S.

```
table(subject$race,subject$biological_sex)
```

	Female	Male
American Indian/Alaska Native	0	1
Asian	32	12
Black or African American	2	3
More Than One Race	15	4
Native Hawaiian or Other Pacific Islander	1	1
Unknown or Not Reported	14	7
White	48	32

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

```
library(lubridate)
```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

```
date, intersect, setdiff, union
```

```
subject$age <- today() - ymd(subject$year_of_birth)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
ap <- subject %>% filter(infancy_vac == "aP")
round( summary( time_length( ap$age, "years" ) ) )
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	27	28	28	29	35

Q8. Determine the age of all individuals at time of boost?

```
library(lubridate)
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

The breakdown is not very representative of the U.S. - this is a serious caveat for the study. However, it is still the largest sample of its type ever assembled.

```
specimen <- read_json("http://cmi-pb.org/api/v5_1/specimen",simplifyVector = TRUE)
ab_titer <- read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer",simplifyVector = TRUE)
head(specimen)
```

	specimen_id	subject_id	actual_day_relative_to_boost
1	1	1	-3
2	2	1	1
3	3	1	3
4	4	1	7
5	5	1	11

	6	1		32
	planned_day_relative_to_boost	specimen_type	visit	
1	0	Blood	1	
2	1	Blood	2	
3	3	Blood	3	
4	7	Blood	4	
5	14	Blood	5	
6	30	Blood	6	

Q9. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:

Merging tables

We need to “join” or link these tables with the ‘subject’ table so we can begin to analyze this data and know who a give Ab sample was collected and when.

```
library(dplyr)

meta<-inner_join(subject,specimen)
```

Joining with `by = join\_by(subject\_id)`

```
head(meta)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	1	wP	Female	Not Hispanic or Latino	White
3	1	wP	Female	Not Hispanic or Latino	White
4	1	wP	Female	Not Hispanic or Latino	White
5	1	wP	Female	Not Hispanic or Latino	White
6	1	wP	Female	Not Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset	age	specimen_id
1	1986-01-01	2016-09-12	2020_dataset	14581 days	1
2	1986-01-01	2016-09-12	2020_dataset	14581 days	2
3	1986-01-01	2016-09-12	2020_dataset	14581 days	3
4	1986-01-01	2016-09-12	2020_dataset	14581 days	4
5	1986-01-01	2016-09-12	2020_dataset	14581 days	5
6	1986-01-01	2016-09-12	2020_dataset	14581 days	6

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	0	0	Blood
2	1	1	Blood
3	3	3	Blood
4	7	7	Blood
5	14	14	Blood
6	30	30	Blood

1	-3	0	Blood
2	1	1	Blood
3	3	3	Blood
4	7	7	Blood
5	11	14	Blood
6	32	30	Blood

visit	
1	1
2	2
3	3
4	4
5	5
6	6

Now let's join the 'ab_titer' table with our 'meta' table so we have all information about a given Ab measurement.

```
ab_data<- inner_join(meta,ab_titer)
```

Joining with `by = join\_by(specimen\_id)`

```
head(ab_data)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female Not Hispanic or Latino	White	
2	1	wP	Female Not Hispanic or Latino	White	
3	1	wP	Female Not Hispanic or Latino	White	
4	1	wP	Female Not Hispanic or Latino	White	
5	1	wP	Female Not Hispanic or Latino	White	
6	1	wP	Female Not Hispanic or Latino	White	

	year_of_birth	date_of_boost	dataset	age	specimen_id
1	1986-01-01	2016-09-12	2020_dataset	14581 days	1
2	1986-01-01	2016-09-12	2020_dataset	14581 days	1
3	1986-01-01	2016-09-12	2020_dataset	14581 days	1
4	1986-01-01	2016-09-12	2020_dataset	14581 days	1
5	1986-01-01	2016-09-12	2020_dataset	14581 days	1
6	1986-01-01	2016-09-12	2020_dataset	14581 days	1

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3	0	Blood
2	-3	0	Blood
3	-3	0	Blood

4			-3		0	Blood
5			-3		0	Blood
6			-3		0	Blood

	visit	isotype	is_antigen_specific	antigen	MFI	MFI_normalised	unit
1	1	IgE	FALSE	Total	1110.21154	2.493425	UG/ML
2	1	IgE	FALSE	Total	2708.91616	2.493425	IU/ML
3	1	IgG	TRUE	PT	68.56614	3.736992	IU/ML
4	1	IgG	TRUE	PRN	332.12718	2.602350	IU/ML
5	1	IgG	TRUE	FHA	1887.12263	34.050956	IU/ML
6	1	IgE	TRUE	ACT	0.10000	1.000000	IU/ML

	lower_limit_of_detection
1	2.096133
2	29.170000
3	0.530000
4	6.205949
5	4.679535
6	2.816431

Q. How many Ab measurements do we have in total

```
nrow(ab_data)
```

```
[1] 61956
```

Q. How many different isotypes (types of Ab) are in the dataset

```
unique(ab_data$isotype)
```

```
[1] "IgE" "IgG" "IgG1" "IgG2" "IgG3" "IgG4"
```

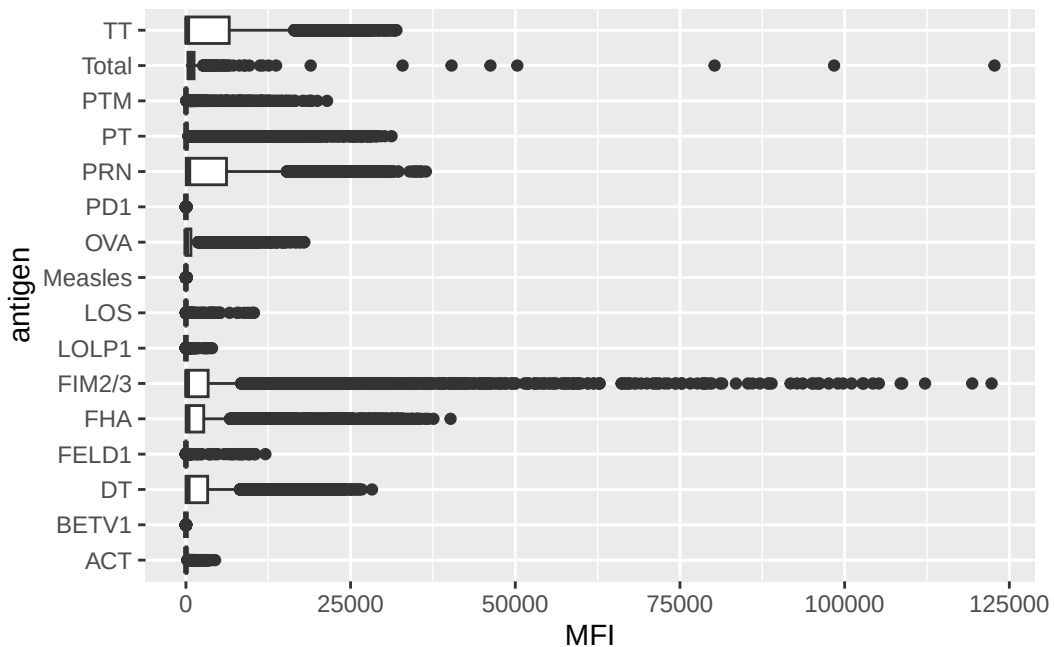
Q. How many different antigens?

```
unique(ab_data$antigen)
```

```
[1] "Total" "PT" "PRN" "FHA" "ACT" "LOS" "FELD1"
[8] "BETV1" "LOLP1" "Measles" "PTM" "FIM2/3" "TT" "DT"
[15] "OVA" "PD1"
```

```
ggplot(ab_data)+
  aes(MFI,antigen)+
  geom_boxplot()
```

Warning: Removed 1 row containing non-finite outside the scale range (``stat_boxplot()``).



IgG is crucial for long-term immunity and responding to bacterial & viral infections

```
igg <- ab_data %>% filter(isotype == "IgG")
head(igg)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	1	wP	Female	Not Hispanic or Latino	White
3	1	wP	Female	Not Hispanic or Latino	White
4	1	wP	Female	Not Hispanic or Latino	White
5	1	wP	Female	Not Hispanic or Latino	White
6	1	wP	Female	Not Hispanic or Latino	White

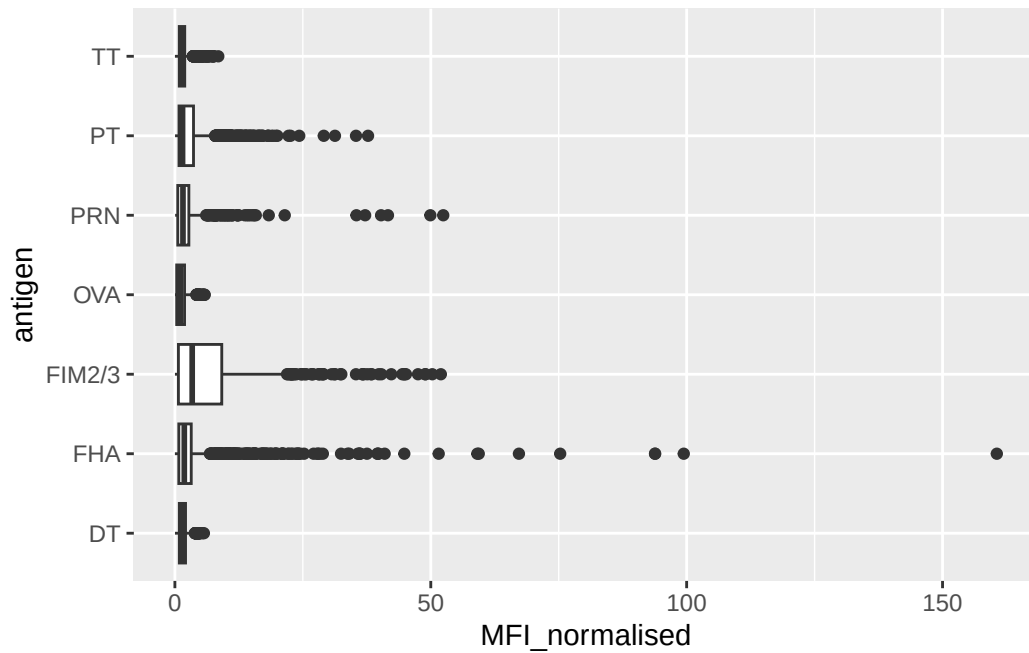
	year_of_birth	date_of_boost	dataset	age	specimen_id
1	1986-01-01	2016-09-12	2020_dataset	14581 days	1
2	1986-01-01	2016-09-12	2020_dataset	14581 days	1
3	1986-01-01	2016-09-12	2020_dataset	14581 days	1
4	1986-01-01	2016-09-12	2020_dataset	14581 days	2
5	1986-01-01	2016-09-12	2020_dataset	14581 days	2
6	1986-01-01	2016-09-12	2020_dataset	14581 days	2

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3	0	Blood
2	-3	0	Blood
3	-3	0	Blood
4	1	1	Blood
5	1	1	Blood
6	1	1	Blood

	visit	isotype	is_antigen_specific	antigen	MFI	MFI_normalised	unit
1	1	IgG	TRUE	PT	68.56614	3.736992	IU/ML
2	1	IgG	TRUE	PRN	332.12718	2.602350	IU/ML
3	1	IgG	TRUE	FHA	1887.12263	34.050956	IU/ML
4	2	IgG	TRUE	PT	41.38442	2.255534	IU/ML
5	2	IgG	TRUE	PRN	174.89761	1.370393	IU/ML
6	2	IgG	TRUE	FHA	246.00957	4.438960	IU/ML

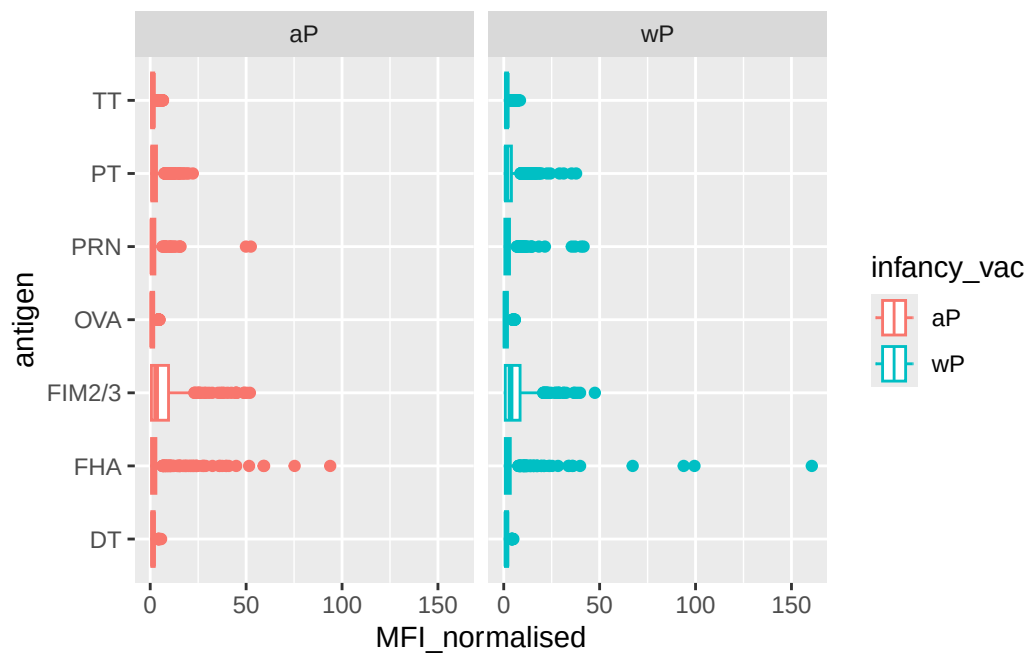
	lower_limit_of_detection
1	0.530000
2	6.205949
3	4.679535
4	0.530000
5	6.205949
6	4.679535

```
ggplot(igg)+
  aes(MFI_normalised,antigen)+
  geom_boxplot()
```



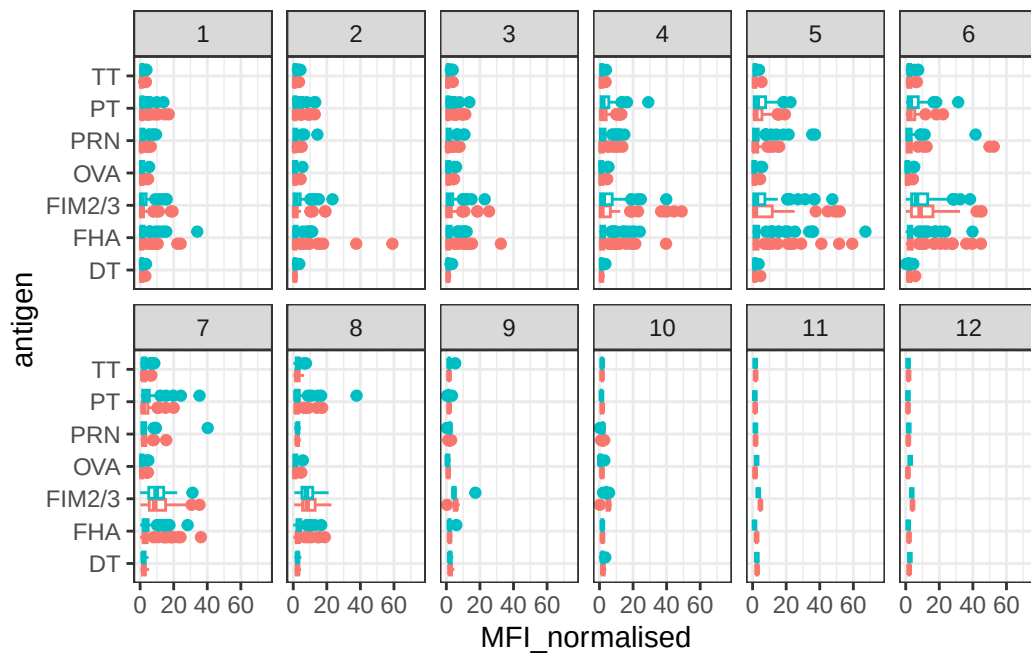
We can “facet” out plot by wP vs aP

```
ggplot(igg)+
  aes(MFI_normalised,antigen,col=infancy_vac)+
  geom_boxplot()+
  facet_wrap(~infancy_vac)
```



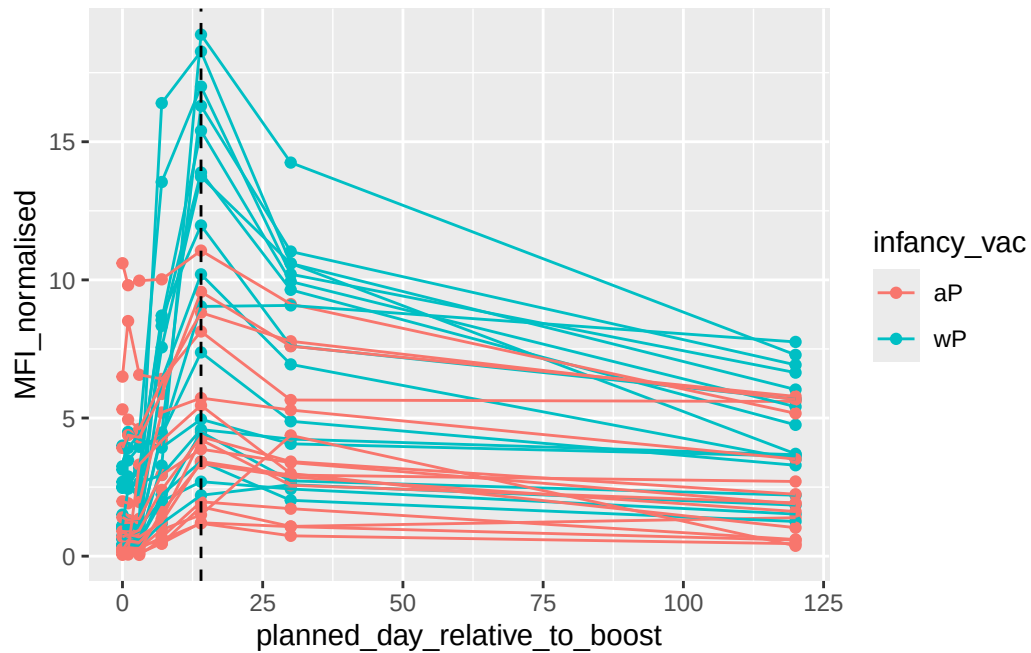
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).



More advanced analysis digging into individual antigen responses over time:

```
filter(igg,antigen=="PT", dataset == "2021_dataset") |>
  ggplot() +
  aes(x=planned_day_relative_to_boost,
      y=MFI_normalised,
      col=infancy_vac,
      group=subject_id) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept=14, linetype="dashed")
```



This plot shows the time course of Pertussis toxin (PT) antibody responses for a large set of wP (teal) and aP (red) individuals. Levels peak at day 14 and are larger in magnitude for wP than aP individuals.

There are lots of cool things to explore in this dataset and we need code and biology knowledge to do it effectively.